

Port Scanning

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Port Scanning

- What is Port :
- Port is the windows that the service work over it
- Let's suppose that you need to start new service , all what you need to open new port

Port Scanning

There are 65535 ports on every IP

These ports can be open , closed , filtered

All what we need to do is to find open ports

Port Scanning

- After finding Open ports , we need to Find what is the services running on these ports ,
- After that we need to know what is the version of these ports

Port Scanning

- We will use tool in linux called nmap

To start nmap

```
# nmap
```

After the word nmap , put the domain name or Ip

```
# nmap google.com
```

```
# nmap 192.168.1.1
```

Port Scanning

After while , you will find open ports appeared as

```
(nossam@nossam) - [~]  
$ nmap google.com  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-07-18 11:43 EEST  
Nmap scan report for google.com (216.58.205.46)  
Host is up (0.056s latency).  
Other addresses for google.com (not scanned): 2a00:1450:4002:416::200e  
rDNS record for 216.58.205.46: mil07s19-in-f14.1e100.net  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
25/tcp    open  smtp  
80/tcp    open  http  
443/tcp   open  https
```


Port Scanning

- We found 3 open ports ,
- 80 , 443, 25
- And every port has a service running on it as :
- Port 80: http
- Port 443: https
- Port 25 : smtp

Port Scanning

- Http , means that there is website running on port 80 , and https means that there is website running on 443
- 25 smtp , means that there is mail server running on port 25 ,
- Mail server to send and receive messages

Port Scanning

But wait does nmap scanned all ports

No , it scans the top 1000 common ports

If we need to scan all ports ,

```
# nmap -p- local.ch
```

Use -p- or -p 65535

Port Scanning

- If you need to check specific port as 80
- `# nmap -p 80 83.247.140.91`
- `# nmap -p80 83.247.140.91`

Port Scanning

There are many sites helps you to find open ports as

search.censys.io

[Shodan.io](https://shodan.io)

You can find Technologies via these sites

Port Scanning

- What we need to know after knowing open ports is
- Services :
- Suppose that found open ports 80,443,25,21
- Then
- `# nmap -p80,443,25,21,22,123 83.247.140.91 -sV`

Port Scanning

```
root@debian-s-2vcpu-8gb-amd-sfo3-01:~# nmap -p80,443,25,21,22,123 83.247.140.91 -sV
```

```
Starting Nmap 7.93 ( https://nmap.org ) at 2024-07-18 09:30 UTC
```

```
Failed to resolve "".
```

```
Failed to resolve "".
```

```
Nmap scan report for 83.247.140.91
```

```
Host is up (0.17s latency).
```

PORT	STATE	SERVICE	VERSION
21/tcp	open	ssh	OpenSSH 6.0p1 Debian 4+deb7u2 (protocol 2.0)
22/tcp	open	ssh	OpenSSH 6.0p1 Debian 4+deb7u2 (protocol 2.0)
25/tcp	filtered	smtp	
80/tcp	closed	http	
123/tcp	closed	ntp	
443/tcp	closed	https	

```
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Port Scanning

- After knowing the version of the service , you can take it copy and go to google and search for exploits
- There are scripts in nmap we are gonna talk about it in advanced level

Port Scanning

- But let's talk about ports ,
- Port 80 means that http or site is running and for us we don't care more about it
- Port 443,8080 : means that https or secure http protocol for the site and don't care about it
- 21 ftp , protocol is used to transfer the files
- (important for us)

Port Scanning

- 22 ssh : is used to control the device remotely
- 25,587 smtp : is mail server running on domain
- 53 DNS : tells that there is dns server running here
- 445,137,139 : Netbios-ssn and SMB , related to windows server and Active directory
- 88 Kerberos : related to authentication server

Port Scanning

- We will know how to exploit common ports and start with ftp
- If you found ftp 21 is open then login with (anonymous:anonymous) as username and password

```
# ftp <ip>
```

```
anonymous and password anonymous
```

If you logged in then vulnerability and you can find all system files

Port Scanning

- `# ftp 84.247.131.64`
- And put username anonymous and password anonymous

Once you logged in then

`# ls`

Port Scanning

- `nmap <ip>`
- `nmap -sV <ip>` ==> to get the version of the service
- `nmap -p22 <ip>` ==> to scan specific port 22
- `nmap -p- <ip>` ==> to scan all ports 65535
- `nmap -f <ip>` ==> to make fragmented scan
- `nmap -Pn <ip>` ==> to bypass firewall (No ping scan)
- `Nmap -sS <ip>` ==> to make Sync scan
- `nmap --script vuln <ip>` => to use script for searching for vulnerabilities
- to find all scripts you can visit `(/usr/share/nmap/scripts)`

Port Scanning

- What is **Masscan**?
- Masscan is a high-performance, open-source port scanner.
- Capable of scanning the entire internet within minutes.
- Designed for fast scanning with minimal system resource usage.

Port Scanning

- Extremely fast scanning speed.
- Supports CIDR, IP ranges, and domain scanning.
- Customizable packet transmission rate.

Port Scanning

- Speed:
- Masscan: Millions of packets per second.
- Nmap: Slower but offers detailed service and vulnerability detection.
- Purpose:
- Masscan: Broad and fast port scanning.
- Nmap: In-depth analysis of specific targets.

Port Scanning

- Installation:
- # `sudo apt update`
- # `sudo apt install git make gcc`
- # `git clone https://github.com/robertdavidgraham/masscan`
- # `cd masscan`
- # `make`

Port Scanning

- How to use it :
- `# masscan [options] <targets>`
- `# masscan -p80 192.168.1.1`
- `# masscan -p22 192.168.1.0/24`

Port Scanning

- Scanning Multiple Ports :
- `# masscan -p80,443,8080 192.168.1.0/24`
- Controls the number of packets sent per second
- `# masscan -p80 192.168.1.0/24 --rate=10000`
- `masscan -p80 192.168.1.0/24 -oX results.xml`

Port Scanning

- Scanning CIDR ranges
- # masscan -p80 192.168.0.0/16
- # masscan -p80 192.168.0.0/16 --exclude 192.168.1.1,192.168.2.1

Port Scanning

- To scan all ip on the internet
- # masscan -p80 0.0.0.0/0 --rate=1000000 --exclude 255.255.255.255

```
root@vmi2314866:/# masscan -p80 0.0.0.0/0 --rate=1000000 --exclude 255.255.255.255
Starting masscan 1.3.9-integration (http://bit.ly/14GZzcT) at 2025-01-04 12:30:33 GMT
Initiating SYN Stealth Scan
Scanning 4294967295 hosts [1 port/host]
^C
```


Port Scanning

- What if I have list of domains , you need to resolve these domains to ip address
 - `# dig -f domains.txt +short >> ip_address.txt`

```
# dig -f domain.txt +short >> ip_address.txt
```


Port Scanning

- And then make command
- `# massdns -r resolvers.txt -o S -w resolved_ips.txt domains.txt` 
- `# masscan -p80 -iL resolved_ips.txt`

Port Scanning

- `# masscan -p80 -iL ip_address.txt`

```
# masscan -p80 -iL ip_address.txt
```